

My Class Notes

GRAVITATION

Newton's Law
of gravitation

Intensity
& Potential

Gr. Field due to
shell, sphere

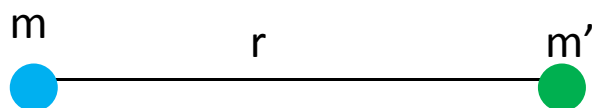
Variation of
 g

Escape Velocity

Satellite
& Parking orbit

Gravitation :

Every particle attracts other particles towards it with a force known as gravitational Force.



Let us consider two point masses m & m' separated by a distance r .

Let F = Gravitational force of attraction between the point masses.

According to Newton :

$$F \propto mm' \longrightarrow (1) \text{ when } r \text{ is constant}$$

$$F \propto \frac{1}{r^2} \longrightarrow (2) \text{ when } m, m' \text{ are constant} \\ \text{– Inverse square law}$$

$$F = G \frac{mm'}{r^2}$$

When G is constant of proportionality and is known as Gravitational Constant.

$$[G] = \frac{[F][r^2]}{[m][m]} = \frac{MLT^{-2}L^2}{MM} = [M^{-1}L^3T^{-2}]$$

$$Unit = gm^{-1}cm^3sec^{-2}, kg^{-1}m^3sec^{-2}$$

The value of $G = 6.667 \times 10^{-11} \quad kg^{-1}m^3sec^{-2}$

Gravitational field :

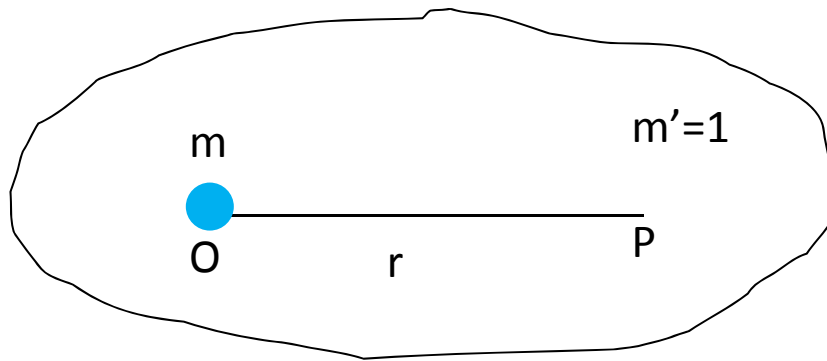
When one body exerts force on another from a distance without any actual contact the force is known as field force.

The region surrounding a mass where if another mass is kept it experiences a force then that region is said to be the gravitational field of the given mass.

The strength of the gravitational field at a point can be measured or represented in two different ways.

- (1) Gravitational Intensity
- (2) Gravitational Potential

Gravitational Intensity : The gravitational intensity at a point in the gravitational field is defined as the force experienced by a unit mass (imaginary) kept at that point.



M = mass of the particle kept at O
P is a point in the gravitational field
of a given point mass at O.

Let us imagine a particle of mass $m'=1$ at P

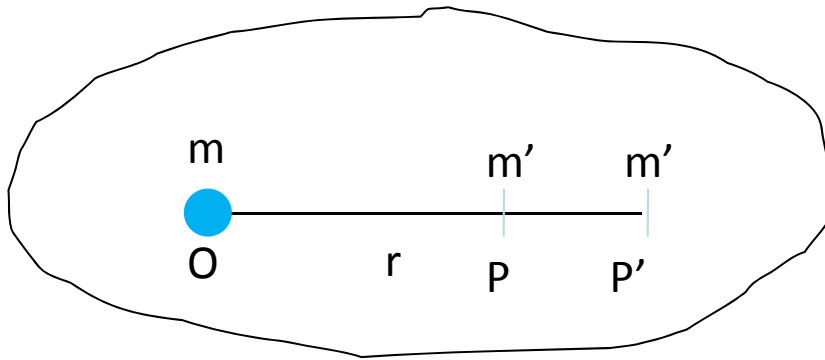
Force experienced by that unit mass at P $= G \frac{m \cdot 1}{r^2}$ along PO and this is known as Intensity at P.

$I = G \frac{m}{r^2}$ It is being a force a vector quantity & is always directed toward the given point mass.

Unit of Intensity $= \frac{\text{Force}}{\text{mass}} = \frac{\text{Newton}}{\text{Kg}}$

Gravitational Potential :

A point mass m is kept at O another point mass m' is considered at P in the gravitational field of the mass at O .



The mass m' experiences a force. If we move the mass from P to P' a displacement occurs in presence of force & work is done energy is spent. This work done is stored in that point mass m' at the point P' in the form of energy known as gravitational potential energy.

The reference point from which the particle is to be moved is taken at infinity.

The work done to bring a particle from infinity up to the given point in the gravitational field is stored in that particle in the form of Gravitational Potential Energy at that point.

The reference point from which the particle is to be moved is taken at infinity.

“ The work done to bring a particle from infinity up to the given point in the gravitational field is stored in that particle in the form of Gravitational Energy at that point.”

Potential : The work done to bring a particle of unit mass from infinity up to the given point in the gravitational field is known as Gravitational Potential at that point.

The Gravitational Potential Energy of a particle per unit mass at point is known as Gravitational Potential at that point.

$$\frac{\text{Work}}{\text{mass}} = \frac{\text{Potential Energy}}{\text{mass}} = \text{Potential}$$

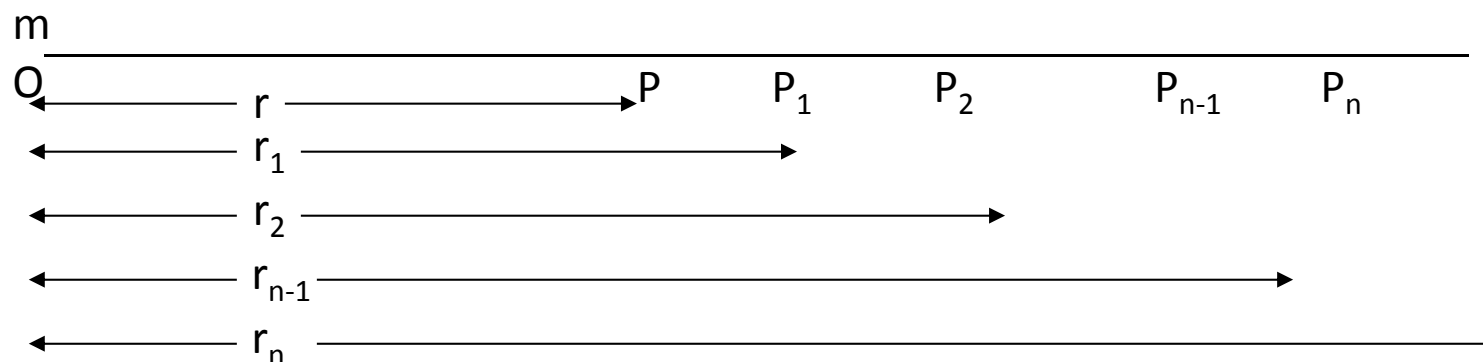
Potential Energy = Potential X mass

Potential Energy : The energy possessed by a body due to its position in a force field is known as potential energy.

Potential energy is calculated from the work done to bring the body from infinity up to the given point in the force field.

Gravitational Potential at a point due to a point mass :

Point mass : If the size of a particle is negligible compared to its distance from the axis (given point) it is known as point mass.



A point mass m is kept at O . P is the given point in the gravitational field of that point mass. Given $OP=r$, join OP and produce it up to any point P_n ($OP_n = r_n$)

Divide PP_n by points $P_1, P_2, \dots, P_{n-1}$ at distances $r_1, r_2, \dots, r_{n-1}$ from O respectively.

$$\text{Gravitational Force on unit mass kept at } P = G \frac{m \cdot 1}{r^2} \quad \text{along } PO$$

$$\text{Gravitational Force on unit mass kept at } P_1 = G \frac{m}{r_1^2} \quad \text{along } P_1O$$

The force changes from P to P_1 . Hence the average force on unit mass between the points P & P_1 can be obtained by taking geometric mean of the two forces because the variation in force is due to the inverse square distance.

The average gravitational force on unit mass considered between the points P & P_1

$$= \sqrt{\frac{Gm}{r^2} \frac{Gm}{r_1^2}}$$

$$= \frac{Gm}{r_1 r} \quad \text{along } P_1 P$$

The work done in taking the unit mass from P to P_1 against the gravitational force
= Force on unit mass $\times$ distance PP_1

$$= \frac{Gm}{r_1 r} (r_1 - r)$$

$$= Gm \left[\frac{r_1}{r r_1} - \frac{r}{r r_1} \right]$$

$$= Gm \left[\frac{1}{r} - \frac{1}{r_1} \right]$$

$$W_{P \rightarrow P_1} = Gm \left[\frac{1}{r} - \frac{1}{r_1} \right]$$

$$W_{P_1 \rightarrow P_2} = Gm \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$W_{P_{n-1} \rightarrow P_n} = Gm \left[\frac{1}{r_{n-1}} - \frac{1}{r_n} \right]$$

Hence the total work done in taking the unit mass from P to P_n can be obtained by taking the algebraic sum of all these work done.

$$W_{P \rightarrow P_n} = W_{P \rightarrow P_1} + W_{P_1 \rightarrow P_2} + W_{P_2 \rightarrow P_3} + \dots + W_{P_{n-1} \rightarrow P_n}$$

$$W_{P \rightarrow P_n} = Gm \left[\frac{1}{r} - \frac{1}{r_n} \right]$$

$$W_{P \rightarrow \infty} = Gm \left[\frac{1}{r} - \frac{1}{\infty} \right] = \frac{Gm}{r}$$

Hence the work done to bring the unit mass from infinity up to P in favor of the attractive force can be obtained by putting a negative sign.

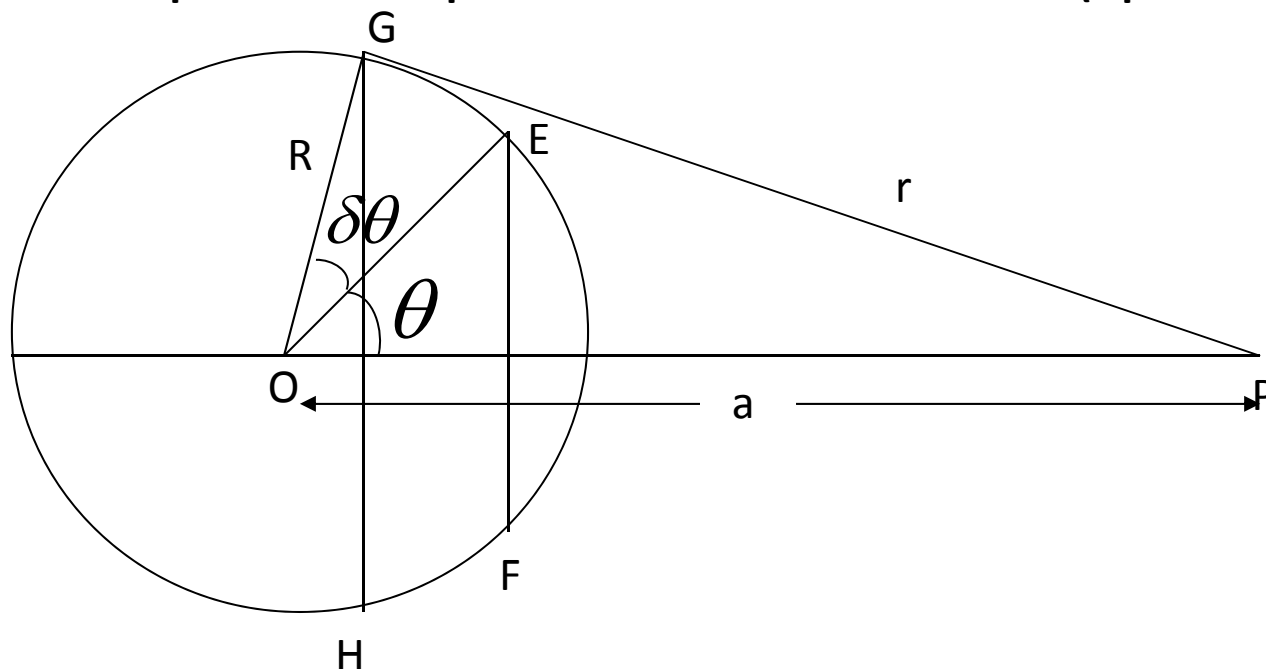
$$W_{\infty \rightarrow P} = - \frac{Gm}{r} \longrightarrow (2)$$

Differentiating equation (2) with respect to r

$$\frac{dv}{dr} = -Gm \frac{d}{dr} \left(\frac{1}{r} \right) = -Gm \left(-\frac{1}{r^2} \right) = \frac{Gm}{r^2} = I$$

The gravitational intensity is rate of change of gravitational potential with distance i.e. potential gradient.

Gravitational potential at a point due to a thin hollow shell (spherical) :



In the figure the circle represents the hollow shell P is the given point.

Given :

M & R = mass and radius of the hollow shell

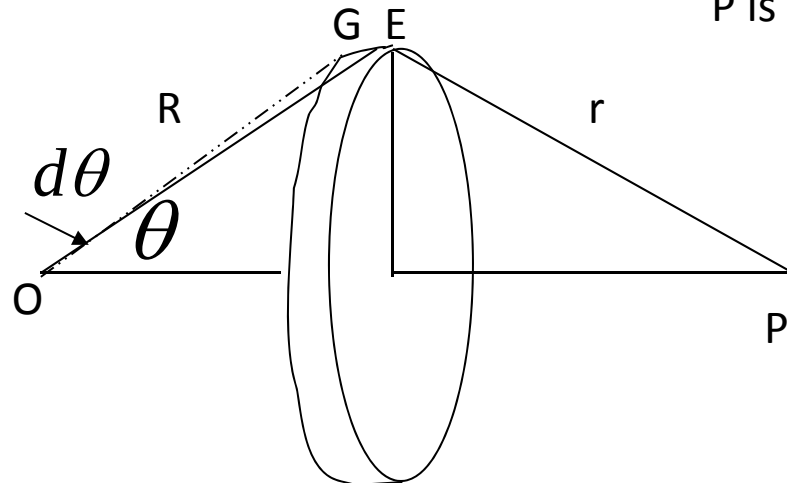
a = OP = distance of the given point P from the center of the shell.

Surface area of the shell $= 4\pi R^2$

Mass per unit area of the shell $= \frac{M}{4\pi R^2}$

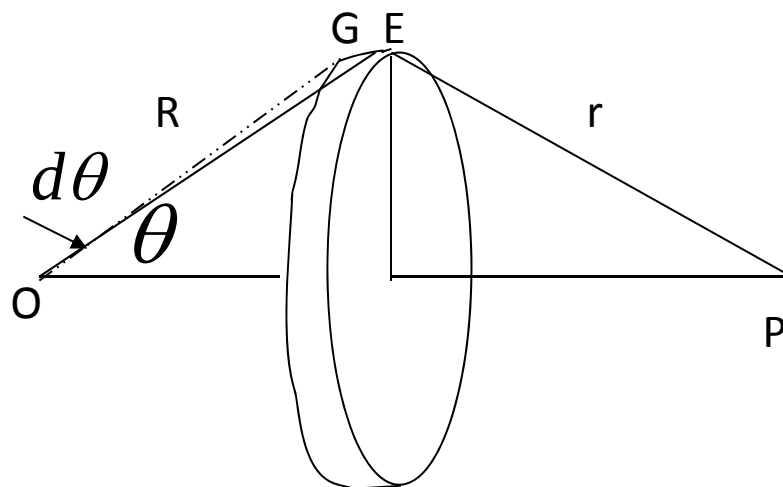
Join OP let us cut the shell by two imaginary planes EF & GH perpendicular to the line OP and separated by a vanishingly small distance.

The cut section appears like an elementary ring & P is a point on the axis of the ring.



Join EP let EP = r, distance of the point P will be same equal to r from every point on the circumference of the ring.

$$\angle POE = \theta, \angle EOG = d\theta$$



From the figure the radius of the elementary ring $= R \sin \theta$

The thickness of the elementary ring $EG = R d\theta$

The surface area of this elementary ring = circumference $\times$ thickness

$$= 2\pi R \sin \theta \times R d\theta$$

$$= 2\pi R^2 \sin \theta d\theta$$

Mass of the elementary ring $= \frac{M}{4\pi R^2} \times 2\pi R^2 \sin \theta d\theta$

$$= \frac{M}{2} \sin \theta d\theta$$

Since the given point P is equidistance from every point on the circumference of the ring potential at P due to elementary ring

$$dv = -\frac{G \times (\text{mass})}{r}$$

$$dv = -\frac{GM \sin \theta d\theta}{2r} \longrightarrow (1)$$

Since the given shell can be assumed to be made up of many such elementary rings. Potential at P due to the whole shell can be obtained by integrating equation (1), but since equation (1) contains two variables r and θ hence it is to be represented in terms of one variable.

From ΔEOP using properties of triangle

$$r^2 = R^2 + a^2 - 2Ra \cos \theta \longrightarrow (2)$$

Differentiating equation (2)

$$2rdr = 0 + 0 - 2Ra(-\sin \theta d\theta)$$

$$\frac{\sin \theta d\theta}{r} = \frac{dr}{Ra} \longrightarrow (3)$$

Putting equation (3) in (1)

$$dv = - \frac{GM}{2} \frac{dr}{Ra} \longrightarrow (4)$$

Case I : The given point P lies out side the shell ($a > R$), in this case from the figure we find that the limits of integration $r = (a - R)$ to $r = (a + R)$

Potential at P

$$V = \int dv = \int_{a-R}^{a+R} - \frac{GM}{2Ra} dr$$

$$V = - \frac{GM}{2Ra} [r]_{a-R}^{a+R} = - \frac{GM}{2Ra} [a + R - a + R]$$

$$V = - \frac{GM}{a} \longrightarrow (5)$$

If we imagine a point mass M (equal to the mass of the shell) at O the center of the shell then potential at P.

$$V = - \frac{GM}{a} \longrightarrow (6)$$

Comparing equation (5) and (6) we can say that to find Gravitational potential at an external point due to a thin hollow shell we can assume the mass of the shell to be concentrated at its center.

Case II : The given point P lies on the surface of the shell ($a = R$)

From the figure in this case the limits of integration is $r=0$ to $r = 2R$

Potential at P :

$$V = \int dv = \int_0^{2R} -\frac{GM}{2R \cdot R} dr$$

$$V = -\frac{GM}{R} \longrightarrow (7)$$

The gravitational potential at any point on the circumference of a shell is constant.

Hence Intensity ($I = \frac{dv}{dr}$) at every point on the surface of a shell is zero.

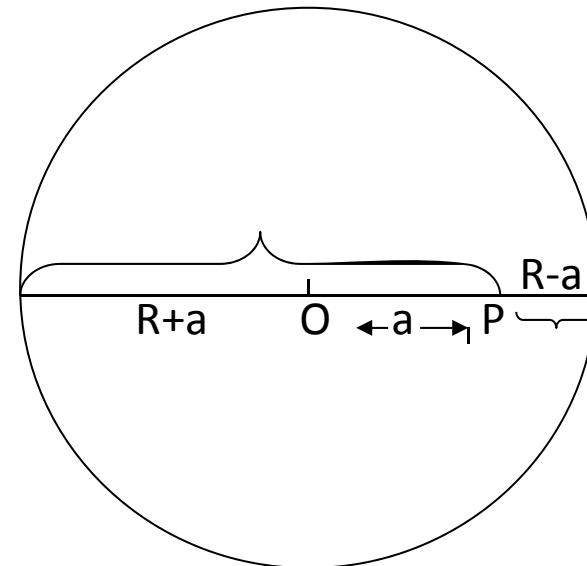
Case III : The given point P lies inside the shell $a < R$

From figure in this case the limits of integration

$R = R-a$ to $r = R+a$

$$V = \int_{R-a}^{R+a} dv = \int_{R-a}^{R+a} -\frac{GM}{2Ra} dr = -\frac{GM}{2Ra} [r]_{R-a}^{R+a}$$

$$V = -\frac{GM}{R} = \text{Constant}$$



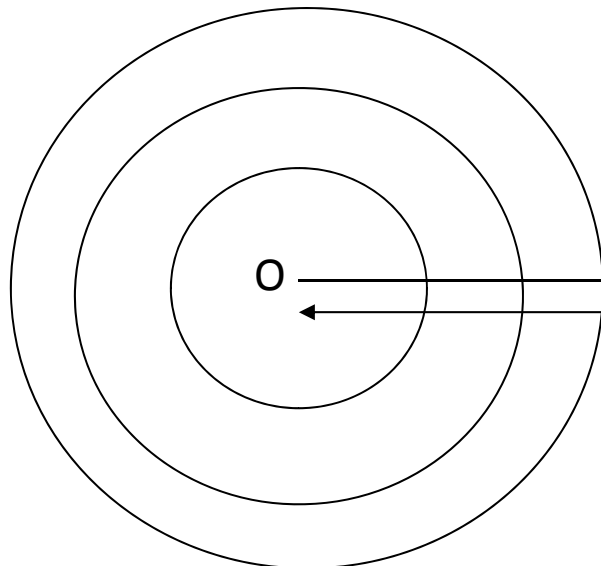
Thus the gravitational potential at any point inside shell is constant & is same as the potential on the surface of the shell. Since the potential is constant the gravitational intensity at any point inside the shell is zero.

Gravitational Potential at a point outside a solid sphere :

Given M & R = Mass & Radius of the solid sphere

OP = a = The distance of the given point from the center of the solid sphere $a > R$

We can imagine the solid sphere to be made up of large no of hollow thin concentric spherical shells of masses $m_1, m_2, m_3, \dots$



Since the given point P lies outside these shell hence potential at P due to these thin shells

$$-\frac{Gm_1}{a}, -\frac{Gm_2}{a}, -\frac{Gm_3}{a}, \dots$$

Hence total potential at P due to these thin shells
i.e. due to the solid sphere

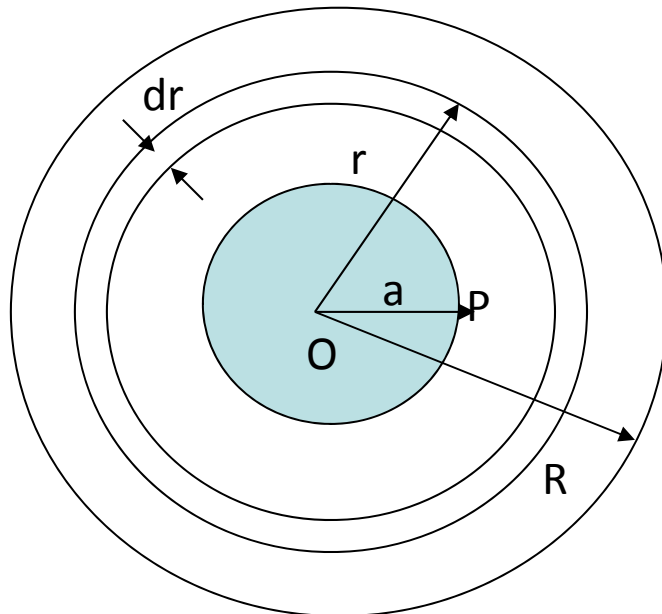
$$V = -\frac{Gm_1}{a} - \frac{Gm_2}{a} - \frac{Gm_3}{a} + \dots = -\frac{G}{a} [m_1 + m_2 + m_3 + \dots] = -\frac{GM}{a}$$

If we assume a point mass M equal to the mass of the solid sphere at the center O of the sphere potential at P

$$V = -\frac{GM}{a}$$

Thus to find the potential at any external point we can imagine the mass of the solid sphere to be concentrated at its center.

Gravitational Potential at a point inside a solid sphere



Given M & R = Mass and Radius of the solid sphere

$OP = a$ = The distance of the given point from the center of the solid sphere ($a < R$)

$$\text{Mass per unit volume} = \frac{3M}{4\pi R^3}$$

Let us imagine that the solid sphere is made up of two parts

- (1) The inner solid sphere with center at O & of radius 'a' passing through the given point P.
(2) The concentric outer thick shell having thickness (R – a)

Let V_1 & V_2 be the potential at the point P due to the inner solid sphere and the outer thick shell respectively.

Total potential at P $V = V_1 + V_2$ $\longrightarrow$ (1)

To find V_1

The volume of the inner solid sphere $= \frac{4}{3} \pi a^3$

Mass of the inner solid sphere $= \frac{3M}{4\pi R^3} \times \frac{4}{3} \pi a^3 = \frac{Ma^3}{R^3}$

Since the given point P lies on the surface of the inner solid sphere the potential at P due to it

$$V_1 = -\frac{G \times \text{mass}}{\text{Radius}} = -\frac{G \times \frac{Ma^3}{R^3}}{a} = -\frac{GMa^2}{R^3} \longrightarrow (2)$$

To find V_2

We can imagine the outer thick shell to be made up of large no of concentric thin shells. Let us consider one such thin shell of radius $r(a \leq r \leq R)$ and radial thickness dr

The volume of the material of the thin shell = Surface area of the shell X thickness
 $= 4\pi R^2 dr$

$$\text{Mass of the elementary shell} = \frac{3M}{4\pi R^3} \times 4\pi r^2 dr = \frac{3M}{R^3} r^2 dr$$

Since the given point P lies inside the thin shell the potential at P due to the thin shell will be same as the potential on the surface of the shell.

Potential at P due to the elementary thin shell

$$dv_2 = -\frac{G \times (\text{mass})}{\text{Radius}}$$

$$dv_2 = -\frac{G \times 3Mr^2 dr / R^3}{r} = -\frac{G3Mr dr}{R^3} \longrightarrow (3)$$

Hence the total potential at P due to the thick shell can be obtained by integrating (3) between the limits $r=a$ to $r=R$

$$V_2 = \int dv_2 = \int_a^R \frac{-G3M}{R^3} r dr = \frac{-G3M}{R^3} \left[\frac{r^2}{2} \right]_a^R$$

$$V_2 = -\frac{GMa^2}{2R^3} [R^2 - a^2] \longrightarrow (4)$$

Putting equation (2) and (4) in (1)

$$V = -\frac{GMa^2}{R^3} - \frac{G3M}{2R^3} (R^2 - a^2)$$

$$V = -\frac{GM}{2R^3} (3R^2 - a^2) \longrightarrow (5)$$

Gravitational field of earth :

Let us consider a particle on or outside the surface of earth.

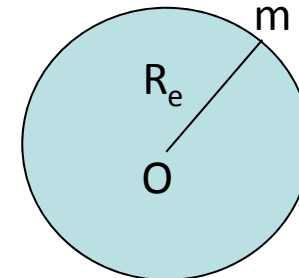
Given M_e and R_e = Mass & Radius of earth respectively

m = mass of the given particle on or outside the surface of earth

Since the given particle lies on or outside the surface of earth to find the force on the particle we can assume the mass of earth (solid sphere) to be concentrated at its center.

Gravitational force with which earth attracts the particle towards its center according to the Newton's law of gravitation

$$F = \frac{GM_e m}{R_e^2} \longrightarrow (1)$$



F = Weight of the particle. This force towards the center of earth produces an acceleration towards the center of earth and since this acceleration is produced by the gravitational force it is known as acceleration due to gravity & is represented by g

$$W = F = mg = \frac{GM_e m}{R_e^2} \longrightarrow (2)$$

$$g = \frac{GM_e}{R_e^2} \longrightarrow (3)$$

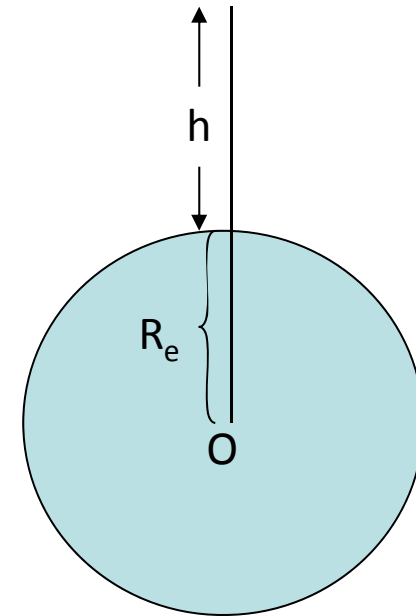
Although from equation (3) g appears to be a constant but we now discuss the factors with which g changes the acceleration due to gravity g varies with following.

- (1) Variation of g with height (altitude)
- (2) Variation of g with depth
- (3) Variation of g with latitude

Variation of g with altitude :

Given M_e and R_e = Mass and Radius of earth

$$g = \frac{GM_e}{R_e^2} \longrightarrow (1)$$



Acceleration due to gravity on the surface of earth. Let us consider a particle of mass m at P at a height h above the surface of earth.

Let g' be the acceleration due to gravity at P . Since the given particle lies outside the surface of earth (solid sphere) assuming the mass of earth to be concentrated at its center, we get the force with which earth attracts the particle towards its center.

$$W = F = \frac{GM_e m}{(R_e + h)^2} = mg'$$

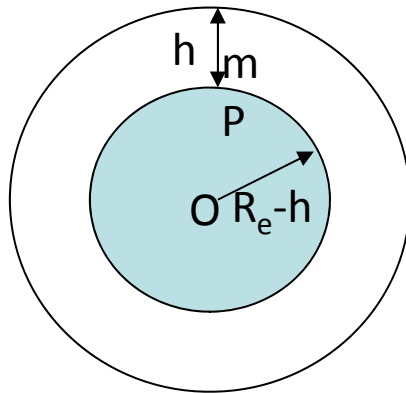
$$g' = \frac{GM_e}{(R_e + h)^2} = \frac{GM_e}{R_e^2 \left(1 + \frac{h}{R_e}\right)^2} = \frac{GM_e}{R_e^2} \left(1 + \frac{h}{R_e}\right)^{-2}$$

$$g' = \frac{GM_e}{R_e^2} \left[1 + (-2) \frac{h}{R_e} + \dots \dots \text{neglected} \right]$$

$$g' = g \left[1 - \frac{2h}{R_e} \right] \longrightarrow (2)$$

Thus we find that acceleration due to gravity decreases as we move above the surface of earth.

(2) Variation of g with depth



Given M_e & R_e = Mass & Radius of earth

$$\rho_e = \frac{3M_e}{4\pi R_e^3}$$

$$g = \frac{GM_e}{R_e^2} = \frac{G \frac{4}{3} \pi R_e^3 \rho_e}{R_e^2}$$

$$g = G \frac{4}{3} \pi R_e \rho_e \longrightarrow (1)$$

Acceleration due to gravity on the surface of earth.

m = mass of the given particle at a depth h below the surface of earth.

Let g' = Acceleration due to gravity at P

Since the given point lies inside the earth (solid sphere) we can not assume the mass of earth to be concentrated at its center.

We can assume the earth to be made up of two parts

(1) The inner solid sphere of center O , Radius ($R_e - h$) passing through the given point P

(2) The concentric thin shell (h is small)

Since the intensity at any internal point of a thin shell is zero hence the force on the particle of mass m due to the outer thin shell is zero. Hence only the inner solid sphere is effective in exerting force on the particle.

$$\text{Mass of the inner solid sphere} = \frac{4}{3} \pi (R_e - h)^3 \rho_e$$

Since the given point P lies on the surface of the inner solid sphere we can assume the mass of it to be concentrated at its center.

Force on the particle

$$F = \frac{G \frac{4}{3} \pi (R_e - h)^3 \rho_e \times m}{(R_e - h)^2} = mg'$$

$$g' = G \frac{4}{3} \pi (R_e - h) \rho_e$$

$$g' = G \frac{4}{3} \pi R_e \left(1 - \frac{h}{R_e} \right)$$

$$g' = g \left(1 - \frac{h}{R_e} \right) \longrightarrow (2)$$

$g' < g$ thus acceleration due to gravity also decreases below the surface of earth.

(3) Variation of g with latitude :

The angle which a particular place on the surface of earth subtends with the equilateral plane at the center of earth is known as latitude of that place.

From the figure the radius of the circle in which the particle rotates

$$r = R_e \cos \phi$$

$$\omega_e = \frac{2\pi}{T}$$

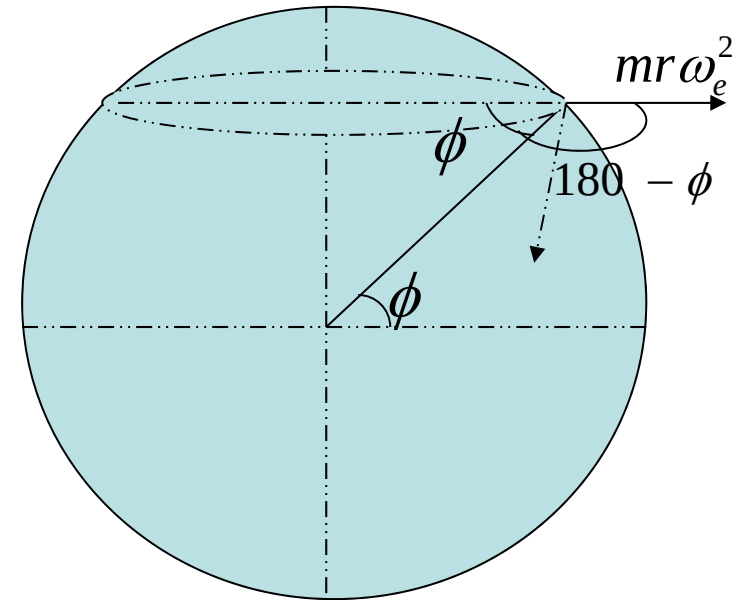
T = Time taken to make one complete rotation about its own axis = 24 hours

Force acting on the particle

(1) Gravitational force of earth towards the center of earth

(2) The centrifugal force $= \frac{mv^2}{r} = \frac{mr^2\omega_e^2}{r} = mr\omega_e^2 = mR_e \cos \phi \omega_e^2$

Using the law of parallelogram of vectors the resultant force on the particle



$$F = mg' = \sqrt{(mg)^2 + (mR_e \cos \phi \omega_e^2)^2} + 2mg \times mR_e \cos \phi \omega_e^2 \times \cos(180 - \phi)$$

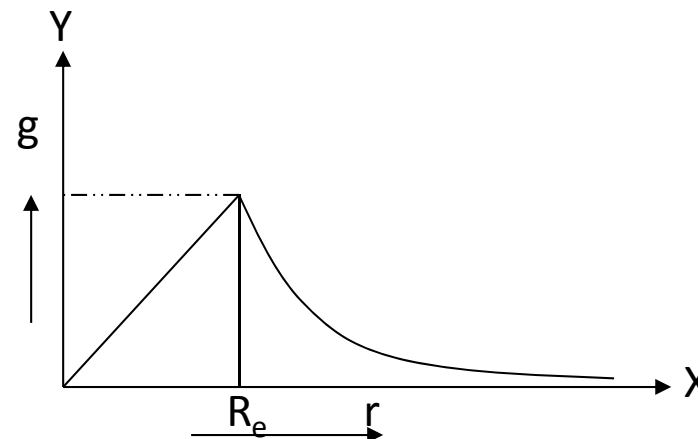
$$mg' = mg \sqrt{1 + \frac{R_e^2 \omega_e^2 \cos \phi}{g^2} - 2 \frac{R_e \omega_e^2 \cos^2 \phi}{g}}$$

$$g' = g \left(1 + \frac{R_e^2 \omega_e^2 \cos \phi}{g^2} - 2 \frac{R_e \omega_e^2 \cos^2 \phi}{g} \right)^{1/2} \longrightarrow (1)$$

From equation (1) we find that since latitude ϕ is different at different places hence g' will also be different at different places.

At the poles $\phi = \pi/2, \cos \phi = \cos \pi/2 = 0$

$g' = g$ Acceleration due to gravity on the surface of earth is maximum at the poles and it is minimum at the equator.

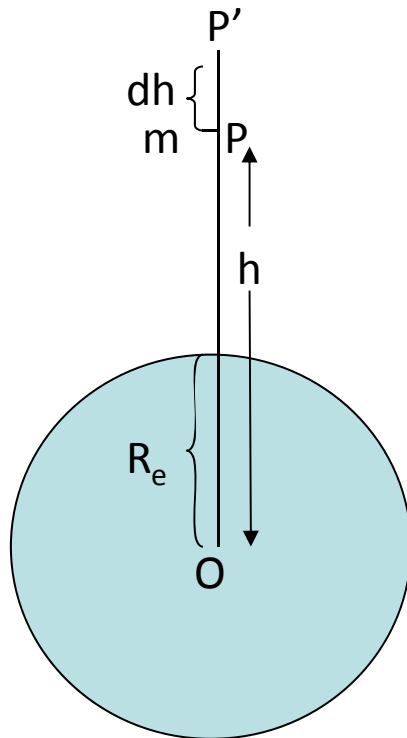


Escape Velocity :

When a particle is thrown up since it moves against the acceleration due to gravity its velocity decreases continuously and when its velocity becomes zero it stops and starts coming down in the direction of acceleration due to gravity. The height up to which it rises depends on the initial velocity of throw.

If a particle can be thrown with such a higher initial velocity that it can go up to infinity i.e. beyond the gravitational field of earth the particle never returns to the surface of earth again and the particle is said to escape the gravitational field of earth.

Definition : The minimum velocity of projection at which any particle go up to infinity i.e. escape the gravitational field of earth and never returns to earth again is known as escape velocity.



Given M_e & R_e = Mass and Radius of earth

$$g = \frac{GM_e}{R_e^2} \longrightarrow (1)$$

Acceleration due to gravity on the surface of earth.

Let us consider a particle of mass m at P at a distance h from the center of earth.

Gravitational force of earth on that particle

$$F = \frac{GM_e m}{h^2} \text{ along } \overrightarrow{PO}$$

To take the particle from P to P' through a small distance dh (so small that the force F between P and P' can be assumed to be constant) against the force work done is dw

$$dw = Fdh = \frac{GM_e m}{h^2} .dh \longrightarrow (2)$$

Hence the total work done in taking the particle from the surface of earth to infinity can be obtained by integrating equation (2)

$$dw = \int dw = \int_{R_e}^{\infty} \frac{GM_e m}{h^2} .dh = GM_e m \left[-\frac{1}{h} \right]_{R_e}^{\infty}$$

$$W = \frac{GM_e m}{R_e} \longrightarrow (3)$$

This work i.e. the energy required to go from the surface of earth to infinity is obtained from the K.E supplied to the particle by throwing it.

Let V_e = Velocity of escape

$$\text{K.E supplied} = \frac{1}{2} m v_e^2 \longrightarrow (4)$$

$$\text{Equating (2) and (3)} \quad \frac{1}{2} m v_e^2 = \frac{GM_e m}{R_e}$$

$$v_e = \sqrt{\frac{2GM_e}{R_e}} \longrightarrow (5)$$

Equation (4) gives the velocity of escape

$$g = \frac{GM_e}{R_e^2}$$

$$gR_e = \frac{GM_e}{R_e} \longrightarrow (6) \quad \text{Putting (6) in (5)}$$

$$v_e = \sqrt{2gR_e} \longrightarrow (7)$$

From (5) and (7) we see that velocity of escape is constant.

$$g = 9.8 \text{ m / sec}^2$$

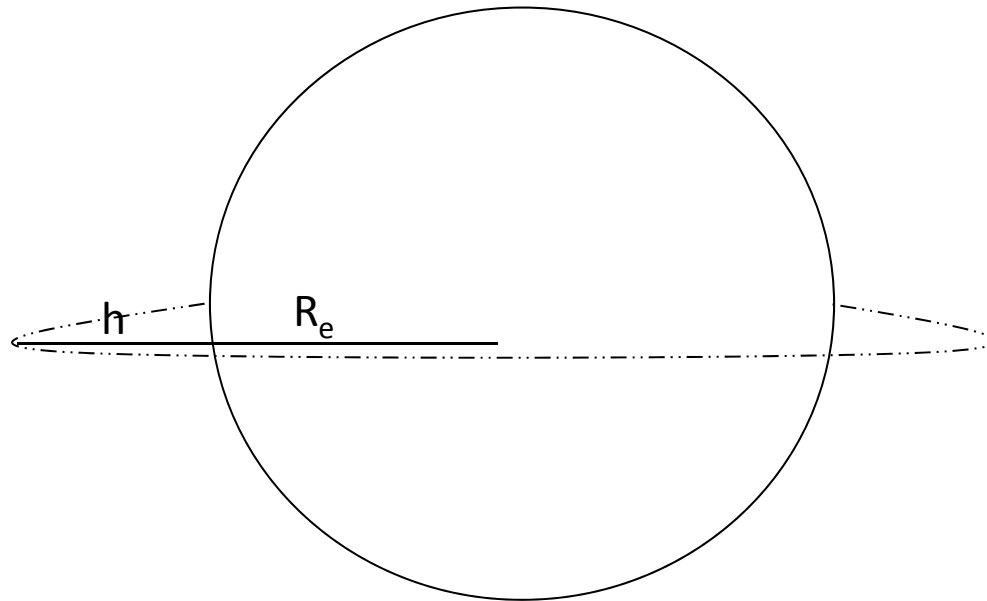
$$R_e = 6400 \text{ KM}$$

$$V_e = 11.2 \text{ KM / sec}$$

Velocity of a satellite :

Any body which revolves around a planet is known as satellite.

We now find the velocity of an artificial satellite of earth.



Given :

M_e & R_e = Mass & Radius of earth

m = mass of the satellite

h = height of the satellite
above the surface of earth.

Let V be the speed of the particle in the circular orbit. The centripetal force required to keep the satellite in uniform circular motion in its orbit is supplied by the gravitational force of attraction of earth.

$$\frac{mv^2}{R_e + h} = \frac{GM_e m}{(R_e + h)^2}$$

$$v = \sqrt{\frac{GM_e}{R_e + h}} \longrightarrow (1)$$

If the orbit of the satellite is close to the surface of earth $h \ll R_e$

The velocity of a satellite close to the surface of earth

$$v = \sqrt{\frac{GM_e}{R_e}} = \sqrt{gR_e}$$

Parking Orbit :

If the relative angular velocity of earth and the satellite is zero i.e. the satellite rotates in such an orbit that the angular velocity of the satellite in that orbit is same as that of earth, then looking from a particular place on the surface of earth the satellite appears to be stationary is known as parking orbit.

Centripetal force = Gravitational force

r = radius of the parking orbit

$\omega = 2\pi / T$ = angular velocity of satellite = ω_e

= angular velocity of earth

$v = r \omega$

$$\frac{mv^2}{r} = \frac{GM_em}{r^2}$$

$$r\omega_e^2 = \frac{GM_e}{r^2}$$

$$r = \left[\frac{GM_e T^2}{4\pi^2} \right]^{1/3}$$

$$G = 6.667 \times 10^{-11} \text{ SI unit}$$

$$T = 24 \times 60 \times 60 \text{ Sec}$$

$$r = 42400 \text{ KM} = (R_e + h)$$

$$r = 42000 - 6400 = 36000 \text{ KM}$$

Thus if the satellite is set in an orbit at a height 36000 KM above the surface of earth the satellite appears to be stationary from the surface of earth.

Weightlessness : The gravitational pull to the earth toward its center is the weight of the body. The pull is zero at the center of earth and at infinite distance from the center of earth. In an artificial satellite the cosmonaut feels weightlessness.

In an artificial satellite the whole of the gravitational pull provides the necessary centripetal force and no part of the gravitational pull remains unbalanced so a body is weightless in an artificial satellite.